

CALCULUS I, Mid-Semester Exam, 11/03/26

Duration: 3 hours, total marks: 60

1. (4+4 points) Compute the following limits:

(a) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + y^2}.$

(b) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2 - x^2 y^2}{\sqrt{x^2 + y^2}}.$

2. (6 points) Let x, y, z be real variables. Suppose that $x^2 y + x z^2 = 5$ and let $w = x^3 y$. Using the chain rule, express the partial derivative $\frac{\partial w}{\partial z}$ of w with respect to z while keeping y constant as a function of x, y, z .

3. (6+2 points) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a differentiable function satisfying $f'(a, b) \cdot (-b, a) = 0$ for all $(a, b) \in \mathbb{R}^2$. Here the symbol $\cdot$ represents the dot product of two vectors in $\mathbb{R}^2$.

(a) Show that f is constant along any circle in $\mathbb{R}^2$ whose centre is the origin.

(b) Determine True or False: f must be constant on all of $\mathbb{R}^2$.

4. (12 points) Define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \begin{cases} \frac{xy(x^2 - y^2)}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

(a) Show that $D_2 f(x, 0) = x$ for all $x \in \mathbb{R}$ and $D_1 f(0, y) = -y$ for all $y \in \mathbb{R}$.

(b) Is f differentiable at $(0, 0)$?

(c) Show that $D_{1,2} f(0, 0) \neq D_{2,1} f(0, 0)$.

5. (12 points) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a differentiable function. Suppose that there is a function $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$f'(a, b) = \begin{bmatrix} \phi(a, b) & 0 \\ 0 & \phi(a, b) \end{bmatrix} \quad \text{for all } a, b \in \mathbb{R}.$$

If $f(0, 0) = (1, 2)$ and $f(1, 0) = (3, 2)$, find $f(5, -5)$.

Hint: If $f = (f_1, f_2)$, you can describe f_1, f_2 explicitly using the given information.

6. (14 points) Let $f = (f_1, f_2) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a function.

Definition 1: f is said to be *complex-differentiable* at $(x_0, y_0) \in \mathbb{R}^2$ if f is differentiable at (x_0, y_0) and the derivative $f'(x_0, y_0) : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is **complex-linear** when $\mathbb{R}^2$ is identified with $\mathbb{C}$ by the map $(a, b) \mapsto a + ib$.

Definition 2: We say that f satisfies the *Cauchy-Riemann equations* if the partial derivatives of f_1, f_2 exist on $\mathbb{R}^2$ and the equations $D_1 f_1 = D_2 f_2$ and $D_2 f_1 = -D_1 f_2$ hold on $\mathbb{R}^2$.

(a) Give an example of a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ which is differentiable but not complex-differentiable on $\mathbb{R}^2$.

(b) If f is complex differentiable on $\mathbb{R}^2$, show that f satisfies the Cauchy-Riemann equations.

(c) Suppose that f satisfies the Cauchy-Riemann equations and the partial derivatives of f_1, f_2 exist and are continuous on $\mathbb{R}^2$. Show that f is complex-differentiable on $\mathbb{R}^2$.